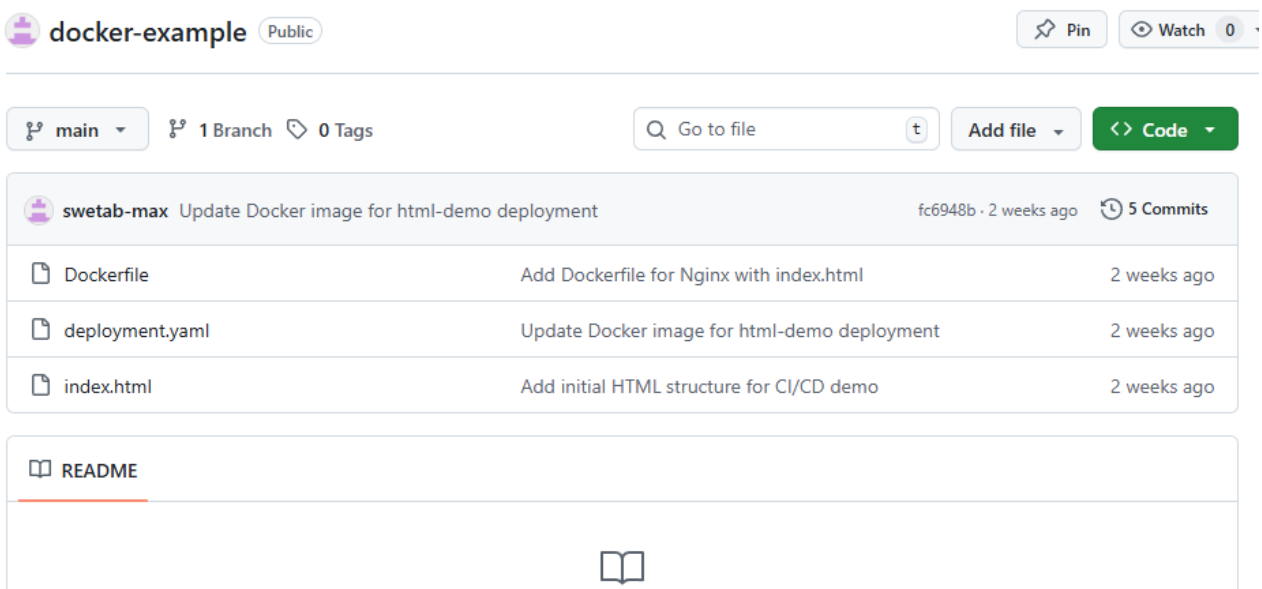
Assignment:

Create an HTML file for your profile and complete the following:

1. Create a repository in GIT and create the following files:

- **index.html**
- **Dockerfile**
- **Deployment.yaml**



The code for the three files are given below:

Create index.html:

```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center;
font-family:Times New Roman;
background-color:#af0d0d;
}
h1{
color:#0ef777;
}
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
```

```
</body>
</html>
```

Create a Dockerfile

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

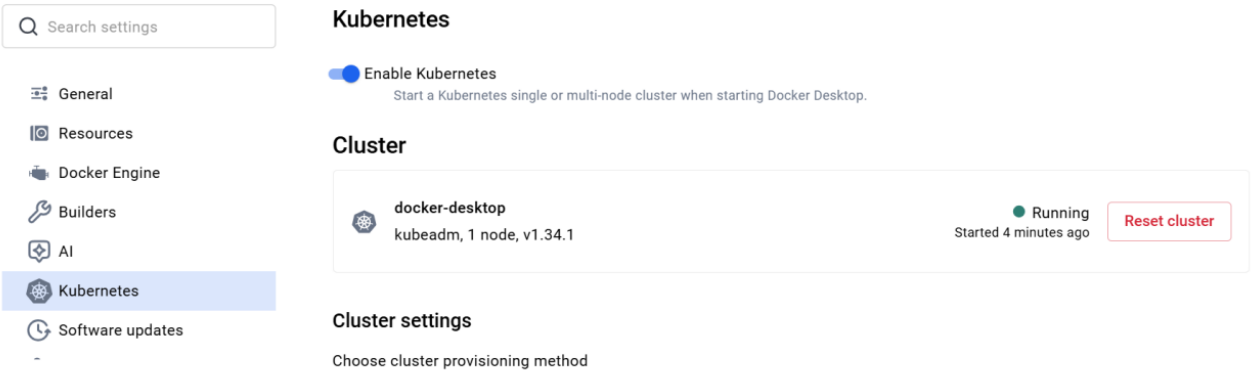
deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
      - name: html-demo
        image: swetab84/html-demo:latest
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30010
```

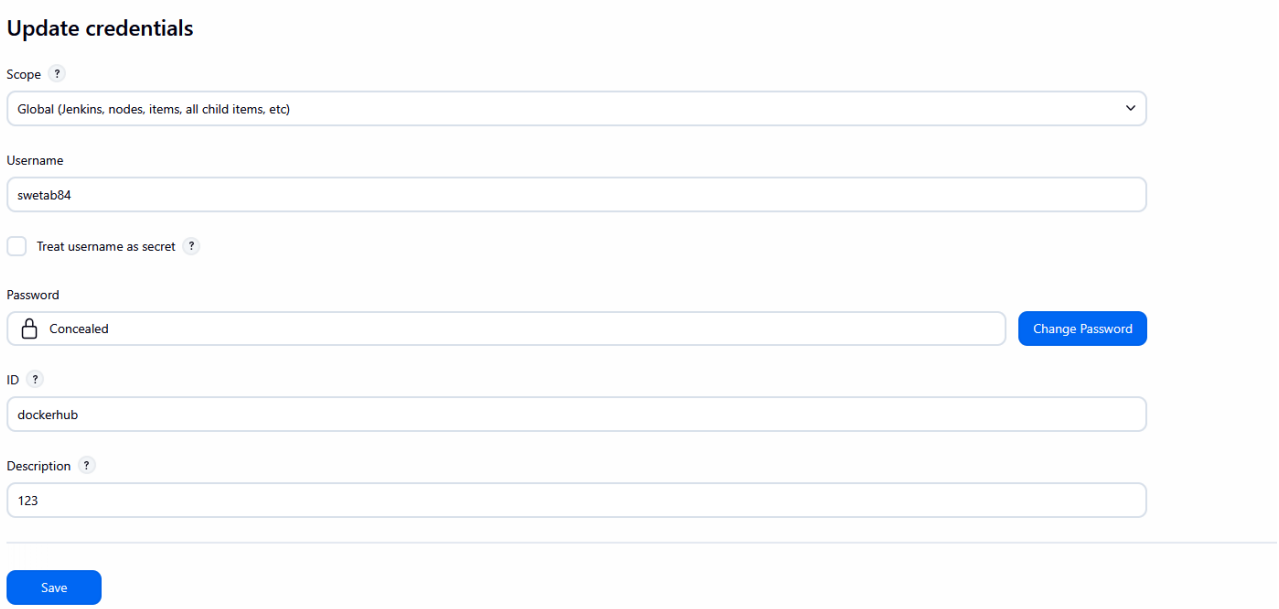
2. Enable Kubernetes
   - **Open the Docker Dashboard**: Launch the Docker Desktop application on your system.
   - **Navigate to Settings**: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.
   - **Select the Kubernetes Tab**: In the Settings menu, find and select the **Kubernetes** tab.
   - **Enable Kubernetes**: Check the box labeled **Enable Kubernetes**.

- **Apply & Restart**: Click the **Apply & Restart** button at the bottom of the settings window. Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time

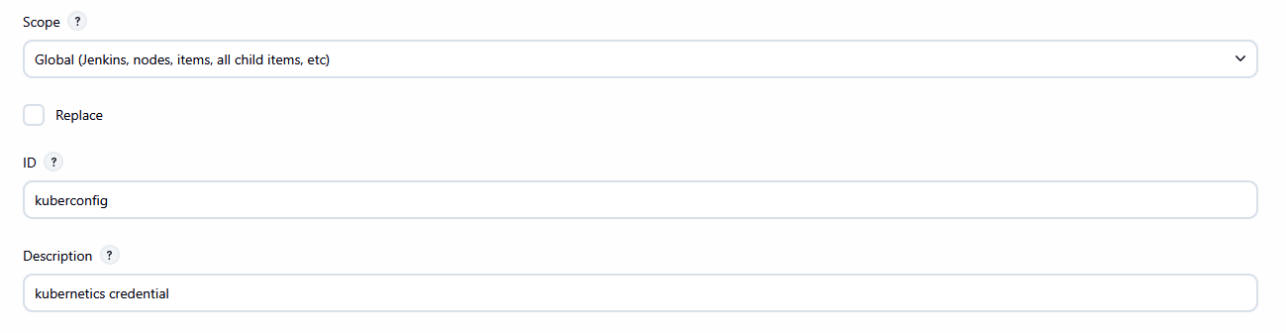


3. Open Jenkins
In Jenkins Setting --> Credentials

New Credentials – User name: <should use the Docker Desktop username>, password <same as Docker Desktop password>, ID:<dockerhub>, Description <any>



New Credentials – Secret – Broswse C: User->Administrator-->.kube



Create New Item -> Pipeline

Paste the code in pipeline script

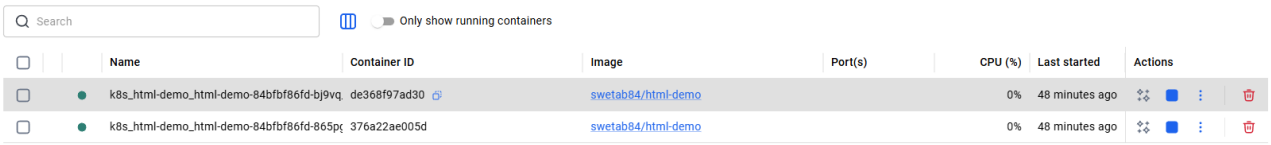
- **Update URL of GIT repository**
- **Make sure credentialsId: “dockerhub” same as mentioned in credentials**
- **Make sure that the ID mentioned in credentials “kuberconfig” is same as in script**
- **Run the Build**

Pipeline:

```
pipeline {
agent any
environment {
DOCKER_IMAGE = "swetab84/html-demo"
}
stages {
   stage('Clone Code') {
steps {
git branch: 'main',
url: 'https://github.com/swetab-max/docker-example.git'
}
}
stage('Build Docker Image') {
steps {
bat 'docker build -t %DOCKER_IMAGE% .'
}
}
stage('Push Image') {
steps {
withCredentials([usernamePassword(credentialsId: 'dockerhub',
usernameVariable: 'USER', passwordVariable: 'PASS')]) {
bat 'docker login -u %USER% -p %PASS%'
bat 'docker push %DOCKER_IMAGE%'
}
}
}
stage('Deploy to Kubernetes') {
steps {
withCredentials([file(credentialsId: 'kuberconfig', variable:
'KUBECONFIG')]) {
```

```
bat '''
set KUBECONFIG=%KUBECONFIG%
kubectl apply -f deployment.yaml --validate=false
'''
}
}
}
}
}
```

*** The same should be reflected in Dockerdesktop



Also the HTML page should open in the port mentioned in deployment.yaml file:
nodePort: 30010

So check if the HTML file open in http://localhost:30010